

Technical Knowledge Handbook

A Quick-Reference Guide for AI Chatbot Knowledge Base

JavaScript

A dynamic, high-level scripting language that powers interactivity on the web. Runs in browsers and on servers via Node.js. Supports functions, closures, and asynchronous programming with `async/await`.

HTML

HyperText Markup Language — the standard markup language for structuring web content using elements such as headings, paragraphs, links, images, and forms.

CSS

Cascading Style Sheets control layout, color, and typography. Uses selectors, the box model, Flexbox, and Grid to style HTML elements and build responsive designs.

React

A JavaScript library for building user interfaces from reusable components. Uses a virtual DOM to efficiently update and render changes to the UI.

JSX

JavaScript XML — a syntax extension used in React that lets developers write HTML-like markup directly within JavaScript to describe UI structure.

State

Data managed within a component that determines its behavior and output. Changes to state (e.g., via React's `useState` hook) trigger a re-render.

Props

Short for 'properties' — read-only data passed from a parent component to a child component to configure or customize it.

Node.js

A JavaScript runtime built on Chrome's V8 engine that enables server-side JavaScript execution, ideal for building fast, scalable network applications.

Express.js

A minimal, flexible Node.js web framework used to build APIs and web servers, handling routing, middleware, and HTTP requests and responses.

REST API

An architectural style for web services that uses standard HTTP methods — GET, POST, PUT, DELETE — to perform CRUD operations on resources identified by URLs.

Git

A distributed version control system that tracks changes to code over time, enabling branching, merging, and collaboration across development teams.

GitHub

A cloud-based platform for hosting Git repositories, supporting collaboration through pull requests, issue tracking, and code review workflows.

SDLC

Software Development Life Cycle — a structured process covering planning, design, development, testing, deployment, and maintenance of software systems.

Software Testing

The process of validating software functionality and quality through unit, integration, system, and acceptance testing to catch defects before release.

TypeScript

A statically typed superset of JavaScript that adds compile-time type checking, improving code reliability, readability, and tooling for large applications.

SQL

Structured Query Language — used to define, manage, and query relational databases using commands such as SELECT, INSERT, UPDATE, and DELETE.

APIs

Application Programming Interfaces are sets of rules and protocols that allow different software systems to communicate and exchange data with one another.

AI Chatbots

Software applications that use natural language processing and machine learning to understand user input, simulate conversation, answer questions, and automate tasks.